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① Data Lake: According to Microsoft Azure platform, data lake is defined as a scalable storage system that allows organisations to store structured, semi structured & unstructured data in its raw format

② IBM defines data mart as a subset of data warehouse designed to serve a specific business unit such as marketing finance or sales. Microsoft Azure defined a data mart as a focused data storage system that improve query performance and accessibility for departmental users

3. A foreign key is a column in a table that refers to the primary key in another table. It is a column or columns in a database that (e.g table 1, column a) are linked to a column in a different ~~language~~ table (table 2, column b)

4. A primary key is a column or columns in a database table with values that ~~are~~ uniquely identify each row or record.

5. Database is a digital repository for storing, managing and securing organised collections of data

6. Data warehouse is a system where data is aggregated from various sources into a central data store (data warehouse) optimized for quiring and analysis. It typically uses extract, transform and load (ETL) processes to clean, prepare and organise data for business intelligence (BI) and other analysis.

7. Schema is a formal description of how data is structured and organised. It is also refered as the structure or design of a database or database object such as table, view, index, stored procedure or trigger. In a relational database, the schema defines the tables, the fields in each table, the relationship between the fields and tables, and the grouping of objects within the database

8. Artificial Intelligence (AI) refers to the intergration of machine learning and advanced algorithm into the data management life circle to automate, optimize and enhance analytical capability

9 Prompt Engineering is the systematic craft of designing precise instructions for large language models (LLM) to achieve domain aligned, high quality results in coding, analysis, modeling and more. According to google cloud prompt engineering is defined as the interface between human intent and machine intelligence.

10 Snowflake is a cloud native, saas-based data platform designed for a large scale data storage, processing and analytics. It is unique because it separate storage from computing, allowing organisations to scale these resources independantly & only pay for what they use.

11. Data bricks is a unified data and AI platform know for pioneering the "Lakhouse" architecture

12. Microsoft sql server is a professional grad relational database management system (RDBMS).

It is essentially a piece of software that functions as a container or manager for one or more databases allowing you to store, organise and retrieve structured data efficiently

13. - Github is a website & a cloud platform owned by Microsoft, that host Git projects online
- Git (the engine) is an actual software you download & use on your computer. It is a version control system that tracks every change you make to your files
Gitlab is also a cloud based platform for GIT projects but it is owned by a different company and has more technical, enterprice focus

14. Version control is an important software development practice for tracking and managing changes made to code and other files.

15 Bigquery is a google Cloud is fully mangged, serverless enterprise datawarehouse.

16. Power BI is a business analytic and data visualization platform by microsoft, that converts raw data from various sources into interactive visual insights

17 Locker studio (formely known as google data studio) is a free cloud-base busness intelligence (BI) and data visualization tool provided by gooyle.

References

1. Azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-a-data-lake
2. www.w3schools.com/sql/sql-foreign key.asp
3. www.cockroachlabs.com/blog/what-is-a-foreign-key/
4. www.ibm.com/think/topics/primary-key
5. www.ibm.com/think/topics/database
6. www.ibm.com/think/topics/data-warehouse
7. Support.esri.com/en-us/gis-dictionary/schema
8. www.ibm.com/think/topics/ai-data-management
9. cloud.google.com/discover/what-is-prompt-engineering
10. www.ibm.com/think/topics/prompt-engineering
11. www.advsyscon.com/blog/snowflake-cloud-data-warehouse/
12. www.capterra.co.za/software/148499/databricks
13. learn.microsoft.com/en-us/sql/sql-server/what-is-sql-server-ver17
14. www.bairesdev.com/blog/git-github-and-gitlab-whats-the-difference/
15. About.gitlab.com/topics/version-control/
16. cloud.google.com/bigquery/pricing
17. www.coursera.org/articles/what-is-power-bi
18. Looker Studio - google.com/u/0/navigation/reporting